12.5 Ocular Infections

A number of microorganisms cause infection when introduced into the mucosa of the eye. In general, bacterial eye infections can lead to inflammation, irritation, and discharge, but they vary in severity. Some are typically short-lived, and others are chronic and lead to permanent eye damage. Prevention requires limiting the exposure to contagious pathogens. When infections do occur, prompt treatment with antibiotics can often limit or prevent permanent damage.

The external surfaces of the eye viz. the conjunctiva and cornea are susceptible to infection. These are exposed to external world and are easily accessible to infective agents. Particularly the conjunctiva is susceptible because it is covered with eyelid that provides warm, moist and enclosed environment in which contaminating organisms can quickly establish a focus of infection. However, eyelid and tears protect the external surfaces of the eye, both mechanically and biologically.

Infection of Eyelid

Most common cause of eyelid infection is *Staphylococcus aureus*.

Infection involves lid margins and cause blepharitis.

When the eyelid glands or follicles are affected stye (sticky eye) is seen (Figure 12.14).

Conjunctivitis (inflammation of conjunctiva) Conjunctivitis or pink eye can be caused by many different kinds of viruses and bacteria.

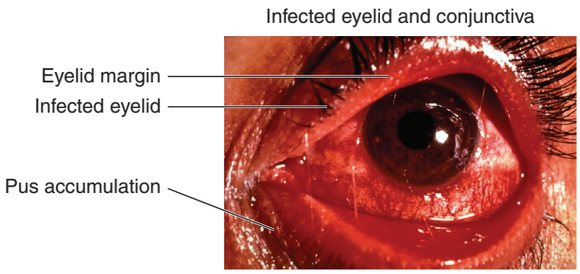


Figure 12.14: Infected eyelid and conjunctiva

Table 12.6: Bacterial infections of the eye

Disease	Pathogen	Signs and Symptoms	Transmission
Acute bacterial conjunctivitis	*Haemophilus influenza*	Inflammation of conjunctiva with purulent discharge	Exposure to secretions from infected individuals
Bacterial keratitis	*Staphylococcus epidermidis*, *Pseudomonas aeruginosa*	Redness and irritation of eye, blurred vision, sensitivity to light; progressive corneal scarring, which can lead to blindness	Exposure to pathogens on contaminated contact lenses
Neonatal conjunctivitis	*Chlamydia trachomatis*, *Neisseria gonorrhoeae*	Inflammation of conjunctiva, purulent discharge, scarring and perforation of cornea; may lead to blindness	Neonate exposed to pathogens in birth canal of mother with chlamydia or gonorrhea

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